

SIOUX FALLS, SD, USA

WMO: 726510

Lat: 43.578N

Lon: 96.754W

Elev: 1428

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-6.3	-17.7	2.2	-10.6	-13.2	2.8	-5.8	32.7	13.0	28.7	15.4	8.0	310	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	91.2	74.2	88.3	73.1	85.6	71.7	77.6	86.8	75.7	85.0	74.0	83.0	13.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	137.9	84.0	72.8	128.5	82.0	71.0	120.7	80.1	42.2	86.9	40.3	85.0	38.6	83.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.1	24.4	20.9	-18.5	96.2	5.5	3.9	-22.5	99.0	-25.8	101.3	-28.9	103.5	-32.9	106.3		
			-18.9	80.7	5.4	1.5	-22.8	81.8	-26.0	82.6	-29.0	83.5	-33.0	84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	16.9	21.2	33.7	46.5	58.2	68.9	73.8	71.0	63.3	49.2	34.5	21.6
	DBStd	22.25	13.02	12.88	13.03	10.33	8.67	6.98	6.09	5.69	8.53	9.59	10.99	11.76
	HDD50	4056	1026	807	525	182	25	0	0	0	6	131	473	881
	HDD65	7442	1491	1226	972	560	247	39	7	13	132	495	914	1346
	CDD50	2851	0	0	18	77	278	567	737	652	405	108	9	0
	CDD65	762	0	0	1	5	36	155	278	199	81	7	0	0
	CDH74	7452	0	0	9	85	423	1556	2820	1706	769	83	1	0
	CDH80	2482	0	0	1	22	115	525	1064	521	217	17	0	0
Wind	WSAvg	10.0	10.1	10.4	11.1	12.0	10.7	9.5	8.8	8.3	9.3	9.9	9.9	9.6
Precipitation	PrecAvg	25.9	0.6	0.7	1.6	2.8	3.4	3.9	2.9	3.2	2.9	2.0	1.1	0.7
	PrecMax	39.2	1.7	4.1	5.0	7.0	8.3	13.7	8.5	9.1	9.3	6.3	4.8	2.6
	PrecMin	11.4	0.1	0.0	0.2	0.2	0.6	0.7	0.2	0.6	0.2	0.0	0.0	0.0
	PrecStd	6.1	0.4	0.7	1.1	1.5	2.0	2.2	2.0	2.0	1.9	1.7	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.9	57.9	73.7	83.3	88.4	94.0	97.2	93.7	89.9	82.3	69.4	52.4
		MCWB	41.2	47.9	55.2	62.1	66.7	73.6	75.8	75.1	71.3	63.7	54.3	44.7
	2%	DB	41.5	49.3	65.8	75.1	83.1	89.2	92.1	88.7	85.7	75.2	62.2	45.7
		MCWB	36.1	42.4	53.3	57.7	65.5	71.9	75.8	75.1	70.3	61.0	51.5	39.6
	5%	DB	37.5	43.1	58.6	69.3	78.9	85.9	88.8	85.4	82.2	69.9	56.0	40.4
		MCWB	34.1	38.0	50.0	54.5	63.6	70.4	74.9	73.0	68.2	58.0	47.5	35.6
	10%	DB	34.4	38.4	52.5	64.2	74.3	82.5	85.8	82.5	78.5	65.2	50.7	36.9
		MCWB	32.0	34.8	45.7	51.6	61.0	68.7	73.2	71.3	66.4	55.4	43.7	33.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	49.9	59.9	64.2	71.3	77.4	80.6	79.4	75.1	67.9	57.5	46.5
		MCDB	49.1	58.0	66.9	78.7	82.7	88.1	89.2	88.3	84.0	75.7	65.6	51.2
	2%	WB	36.9	42.5	54.9	59.8	68.0	74.8	78.3	76.9	72.9	63.5	52.5	39.6
		MCDB	40.9	48.4	63.0	72.5	79.2	85.4	87.5	85.6	82.2	71.2	60.4	45.0
	5%	WB	34.3	38.3	50.5	56.3	65.2	72.7	76.7	74.9	70.7	59.9	48.1	36.3
		MCDB	37.1	42.8	58.6	66.7	75.7	82.7	85.9	83.1	79.1	67.8	55.2	40.0
	10%	WB	32.1	34.8	45.4	53.1	62.5	70.5	74.8	73.0	68.4	56.4	44.0	33.8
		MCDB	34.5	38.1	52.3	62.2	71.9	80.2	84.0	80.4	76.1	64.1	50.7	36.6
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.0	19.0	21.7	21.2	20.4	20.3	19.9	21.9	21.0	19.1	16.5
		MCDBR	21.4	23.9	29.3	30.5	27.1	24.0	22.3	22.3	25.1	28.3	28.3	22.5
		MCWBR	17.0	17.4	18.3	16.6	14.0	11.4	10.5	11.2	12.3	15.9	18.2	16.7
	5% WB	MCDBR	20.2	22.8	27.5	26.4	23.4	20.7	20.1	19.7	20.5	23.7	26.5	21.2
		MCWBR	16.2	16.8	17.6	15.8	13.3	11.3	11.1	11.1	11.8	14.8	17.7	16.3
Clear-Sky Solar Irradiance	taub	0.272	0.285	0.319	0.370	0.388	0.397	0.402	0.405	0.356	0.327	0.294	0.274	
	taud	2.431	2.417	2.406	2.286	2.316	2.348	2.332	2.301	2.415	2.469	2.478	2.440	
	Ebn at Noon	276	292	293	283	280	278	274	269	276	271	265	264	
	Edh at Noon	23	28	33	40	40	39	39	39	32	27	22	21	
All-Sky Solar Radiation	RadAvg	605	939	1227	1547	1781	1998	2118	1749	1395	946	654	485	
	RadStd	53	67	95	139	129	129	78	131	103	111	45	40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page